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Seasonal dynamics of root growth and desiccation cracks and their effects on soil hydraulic conductivity

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ABSTRACT

Vegetation significantly influences soil hydraulic conductivity, with the extent of this influence depending on root morphology and density, which vary across different developmental stages of plants. This research investigates the interaction dynamics between plant roots (during both growth and decay) and desiccation cracks, as well as the combined impact of vegetation, cracks, and seasonal variations on soil hydraulic conductivity (K_{sat}). Root growth and decay patterns were observed using a minirhizotron, while changes in crack formation were monitored and interpreted using the Crack Intensity Factor (CIF) for both vegetated and bare areas over an eighteen-month period of wetting and drying cycles. K_{sat} was analysed based on data from a double-ring test. The findings indicate that the presence of vetiver roots results in a less visible and uneven crack distribution compared to bare soil, with CIF and average crack widths reduced by half. However, cracks reappear during root decay periods. Although cracks were minimised in vegetated soil, K_{sat} values increased significantly during dry periods, with a 16-fold rise in the vegetated zone due to root propagation, while the bare zone showed a marginal 5-fold increase. The presence of cracks and roots significantly influences K_{sat}, exhibiting distinct hysteresis behaviour in response to drying and wetting cycles.

1. Introduction

Soil hydraulic conductivity is crucial for many applications, including evaluating water availability and runoff in agriculture and modelling soil slope stability (Apriyono and Yuliana, 2023; Tsiampousi et al., 2017; Yuliana et al., 2023). With the growing implementation of nature-based solutions, vegetation has been increasingly utilized as a natural material for erosion control and slope stabilisation (Bordoloi and Ng, 2020; Chen et al., 2023; Kamchoom and Leung, 2018). While vegetation can provide soil mechanical reinforcement (Kamchoom et al., 2014; Wu et al., 2021), the root morphology and density, which vary across different developmental stages of the plants, also influence soil hydraulic conductivity (Boldrin et al., 2022; Huat et al., 2006). Research on the effects of vegetation on saturated hydraulic conductivity (K_{sat}) has produced contradictory findings. A laboratory test conducted by Rahardjo et al. (2014) measured K_{sat} in soils containing vetiver (*Chrysopogon zizaniodes*) and orange jasmine (*Murraya exotica L.)* roots and found a decrease in K_{sat} compared to bare soil. They attributed this reduction to the dense root systems creating more tortuous pathways for water flow, thereby reducing permeability. A constant head method in a permeability test apparatus was utilized by Nguyen et al. (2020) in a laboratory study to determine the effect of grass age on K_{sat}. Their results indicated a negative relationship between root biomass and K_{sat}, showing that as root biomass increased with grass age, K_{sat} decreased. Conversely, several studies on soil columns support the idea that vegetation, through the development of root systems, can significantly enhance K_{sat}. Another study conducted by Leung et al. (2018) found that root growth in both *Salix viminalis* and *Lolium perenne x Festuca pratensis* hybrid significantly enhanced K_{sat}. A significant positive correlation between root biomass and K_{sat} in plants of three different ages was observed by Rajamanthri et al. (2021). Similarly, an averagely four-times greater K_{sat} in soil vegetated with herbaceous species compared

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to that in bare soil was reported by Boldrin et al. (2022).

Additionally, the temporal dynamics of root growth and decay can cause variations in root biomechanical properties and soil reinforcement (Kamchoom et al., 2022a; Zhu et al., 2020). These dynamic changes of roots can modify the crack pattern by potentially altering the way cracks propagate through the soil Vegetation can facilitate preferential pathways through primary root growth, and obstructing flow channels as secondary roots develop (Gholami et al., 2024). Dense vegetation leads to discrete, linear cracking styles due to root permeation (Yu et al., 2021). Additionally, the cracks in vegetated soil were relatively higher than those in bare soil throughout the test period was observed by Gadi et al. (2017). These studies collectively support the notion that vegetative growth during moist conditions facilitates the creation of preferential flow paths and enhances water infiltration However, root growth during dry periods presents a contrasting scenario. Vegetation could mitigate desiccation cracking during drought, particularly on red clay slopes was reported by Gao et al. (2021). Their research indicates that vegetation, especially with deeper root systems, helps stabilise the soil by reducing the formation of desiccation cracks, which can otherwise exacerbate soil erosion and instability. In addition, the decay of roots adds another layer of complexity. As roots decay, they leave behind channels and voids that can enhance preferential flow paths. However, decay and growth can occur affecting the total root biomass and soil porosity across different seasons. Based on the aforementioned studies, it appears that most investigations have focused on a single stage of plant growth, potentially overlooking the dynamic influence of changing root content during growth and decay on crack formation and saturated hydraulic conductivity (K_{sat}). Moreover, the impact of seasonal variations on these processes has not been adequately considered in many studies. It is imperative to investigate the interplay between plant root dynamics, soil cracks and saturated hydraulic conductivity, as well as how these factors mutually influence one another throughout different seasons. The lack of systematic field observations, particularly regarding the effects of varying seasons on both root dynamics and crack formation, limits our capacity to establish a comprehensive understanding of soil hydraulic behaviour in vegetated soils throughout the changing seasons.

The study aims to investigate the interaction dynamics between plant roots (during both growth and decay) and desiccation cracks, as well as the combined impact of vegetation, cracks, and seasonal variations on soil hydraulic conductivity. Root growth and decay patterns in the field study were observed using a minirhizotron. Concurrently, corresponding changes in crack formation were monitored using the same approach and interpreted using the CIF for both vegetated and bare areas. These methods enable the comprehensive observation of root dynamics, crack variations, and their intricate interplay over an eighteen-month period of wetting and drying cycles. The K_{sat} was analysed based on data collected from a double-ring test. These findings are expected to yield an important understanding of how vegetation influences the formation and propagation of cracks, ultimately affecting soil hydraulic conductivity throughout diverse seasons. The outcome also reveals the hysteresis behaviour of K_{sat} in vegetated soils, providing critical insights into soil property changes over time and their relevance to geotechnical applications such as slope stabilisation, landfill cover design, and erosion control.

2. Methodology

2.1. Test setup

The field monitoring was conducted at a designated test site in Khlong Sam Wa, Thailand. The test site featured a flat ground area divided into three square zones, as shown in Fig. 1, each measuring 1.5 m by 1.5 m. These dimensions were selected to provide sufficient space for observing root growth, soil porosity changes, and desiccation crack formation over time. Two of the zones were designated as vegetation zones (V1 and V2) and planted with vetiver grass (*Vetiveria zizanioides*), while the third zone in the middle remained bare (B) to serve as a control for comparison. This setup allowed for a detailed assessment of natural variations and the specific impacts of the grass on soil hydraulic conductivity, root dynamics, and desiccation crack development.

Soil samples were collected from the test site to evaluate the index soil properties. The soil was excavated to a depth of 5 m, and observations confirmed a consistent soil profile up to this depth. The Atterberg limits were determined following ASTM D4318 (2014), revealing a liquid limit of 53.61 %, a plastic limit of 14.50 %, and a plasticity index of 39.11 %. These high values indicate significant soil plasticity, which is crucial for understanding soil desiccation crack development under

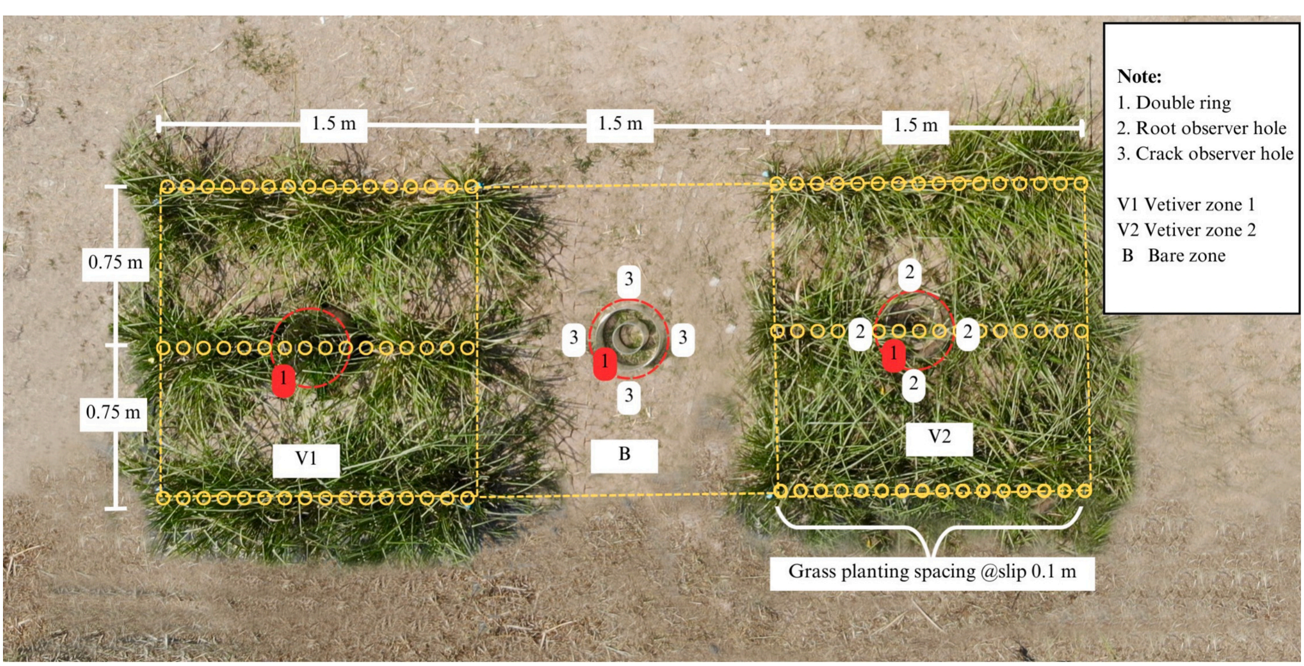

Fig. 1. Overview of the monitoring area showing the spatial division into bare and vegetated zones.

vegetative and seasonal influences. The specific gravity of the soil was measured at 2.57, typical for clay soils. The clay fraction was identified as approximately 51 %, and the soil was classified as high-plasticity clay by the Unified Soil Classification System. Field dry density, measured using the sand cone method (ASTM D1556 (2007)), was 13.56 kN/m^3.

Vetiver grass was selected due to its wide use in soil and water conservation, sediment control, land stabilisation, and ecological restoration (Phan et al., 2022; Phan et al., 2021). Indeed this species has demonstrated to be effective in stabilising slopes, controlling erosion (Stokes et al., 2008a), and mitigating soil expansion (Wang et al., 2020). Vetiver thrives in tropical, semi-tropical, and Mediterranean climates, adapting to poor soil and extreme weather conditions (Troung et al., 2008; Wasino et al., 2019). The planting configuration for vetiver grass in this research was based on recommendations from the Land Development Department (2014) and Troung et al. (2008). Vetiver tillers were planted in single rows with 10 cm of spacing within each row. The distance between rows was set at 75 cm, providing adequate space for root growth and minimising competition between plants. This configuration is adaptable, with typical recommendations suggesting 5–10 cm spacing within rows and 50–100 cm between rows.

2.2. Determination of K_{sat} from infiltration test

In this study, the infiltration rate was measured using the double-ring infiltrometer (DRI) test, a method known for its high accuracy in preserving vertical water infiltration (Verbist et al., 2010). The DRI test minimises lateral flow, ensuring predominantly vertical water movement, which is crucial for accurate infiltration rates (Lai and Ren, 2007; Li et al., 2019). Fig. 2 shows the details of the DRI setup at the field observation. Both vegetated and bare areas were tested using the same DRI setup for consistency. The inner and outer rings of the DRI were made from acrylic tubes with diameters of 15 cm and 30 cm, respectively, similar to previous studies (Gregory et al., 2005). The outer ring reduces errors from lateral water flow, promoting vertical flow beneath the inner ring (ASTM D 3385, 2003; Fatehnia et al., 2016). Transparent acrylic tubes were selected for their ability to allow sunlight to pass through, ensuring minimal disturbance of the natural evapotranspiration process and plant growth. The inner and outer rings were embedded in the soil to depths of 10 cm and 20 cm, respectively. By keeping the setup permanently installed, the integrity of the soil structure was maintained, ensuring that the infiltration measurements were not affected by repeated installation and removal of the equipment.

The DRI test procedures in this study followed ASTM D3385-09 (2016) standards, employing the falling-head method suitable for the site's soil conditions. The water flow rate within the inner ring was meticulously monitored using an HC-SR04 ultrasonic distance sensor, positioned atop the ring, capable of measuring distances from 2 cm to 400 cm with an accuracy of 0.3 cm. This sensor works by emitting an ultrasonic pulse and measuring the time it takes for the pulse to return after reflecting off the water surface (Jeswin et al., 2017). This method ensures precise distance measurements, critical for accurately recording changes in the water level within the inner ring. The sensors were connected to an Arduino Uno datalogger programmed to record water level fluctuations at one-second intervals, allowing detailed monitoring of the infiltration process and providing a robust dataset for analysis. The K_{sat} was determined using the Elrick model (Elrick et al., 1995), which can estimate hydraulic conductivity based on early-time infiltration data under falling-head conditions. This model is particularly effective for low-permeability soils, where achieving full saturation is challenging due to the difficulty of eliminating entrapped air (Gholami et al., 2024). By analysing the change in water level over time, this method offers an indirect but reliable estimate of K_{sat} (Bagarello et al., 2004; Elrick et al., 1995). According to Elrick et al. (1995), infiltration can be described as:

$$I(t) = S.t^{\frac{1}{2}} \quad (1)$$

where I(t) is the cumulative infiltration, S is the soil sorptivity at the ponded head (m $s^{-0.5}$), and t refers to the time (s). White and Sully (1987) proposed an approximate analytical relationship among S, K_{sat}, and ϕ_m, which is expressed as follows:

$$S = \left\{2(\Delta\theta)K_{sat}H + \left[\frac{(\Delta\theta)\phi_m}{b}\right]\right\}^{\frac{1}{2}} \quad (2)$$

where$\Delta\theta$ is the difference between saturated and initial water contents (m^3/ m^3), H is the ponded head (m), ϕ_m is the matrix potential (m^2 s^{-1}), and b is a constant value of 0.55.

The substitution of Eq. (2) into Eq. (1) can be written as:

(a) (b)

Fig. 2. Schematic representation illustrating: (a) elevation and (b) plan view of the double ring test setup.

$$\mathrm{I}(t) = t^{\frac{1}{2}}\left\{2(\Delta\theta)K_{sat}H + \left[\frac{(\Delta\theta)\phi_m}{b}\right]\right\}^{\frac{1}{2}} \tag{3}$$

Under the falling-head condition, the cumulative infiltration is a function of H as follows:

$$\mathrm{I}(t) = \left(\frac{c}{A}\right)(H_0 - H_t) \tag{4}$$

Where c and A are the cross-sectional areas of the falling-head tube and the soil sample, respectively. By substituting Eqs. (2) and (4) into Eq. (1), it can be written as follows:

$$t^{\frac{1}{2}} = \frac{c}{A}\frac{(H_0 - H_t)}{\left\{2(\Delta\theta)K_{sat}H + \left[\frac{(\Delta\theta)\phi_m}{b}\right]\right\}^{\frac{1}{2}}} \tag{5}$$

K_{sat} can be determined using simplified Eq. (5), which is fitted with measured early falling head data through an iterative process involving the parameters $\Delta\theta$ and ϕ_m. This process continues until the value of K_{sat} converges to a stable result, ensuring an accurate calculation of the saturated hydraulic conductivity. The values of $\Delta\theta$ and ϕ_m for each test are depicted in the supplementary Table 1.

2.3. Observation of root area ratio and crack propagation

In this study, the root and crack density were observed directly on the outer ring's side, highlighted in red in Fig. 2. The objective was to understand how root growth and crack formation influence soil hydraulic properties, particularly under seasonal variations. This involved creating observation holes in zone B and V2, capturing images of roots and cracks, and analysing these images using advanced digital processing techniques. Four cube-shaped holes, each measuring 10 cm × 10 cm with a depth of 10 cm. These holes served as observation points for roots and cracks. The 10 cm cube holes were excavated outside the outer ring (see Fig. 2b) to ensure no disturbance to the inner area where the infiltration tests were conducted. Once the holes were no longer in use, they were covered with black plastic sheets to prevent light penetration into the soil. Before the infiltrometer tests, a digital camera was inserted into these holes to capture images of roots and cracks. In the vegetated zone, the camera provided comprehensive views of root and crack interactions. Similarly, in the bare area, the camera captured crack images, allowing comparisons between vegetated and non-vegetated zones.

Root content was quantified using the side root area ratio (RAR), which is calculated as the root area over the total area. RAR serves as an index of root content, supported by a strong correlation between side root area ratio and root biomass per soil volume, as demonstrated by Mahannopkul and Jotisankasa (2019). This metric provides a measure of the proportion of the soil occupied by roots, offering insights into root density and distribution. Similarly, crack concentration was defined using the crack intensity factor (CIF), which is determined by dividing the area of cracks by the total area. This factor quantifies the extent of cracking within a given soil area, allowing for the assessment of crack development and propagation in soil. This study highlights the importance of examining the crack profile along the soil depth, which is directly influenced by root dynamics, including growth and decay. These cracks also play a critical role in pore connectivity and continuity, significantly affecting hydraulic conductivity and water movement along the soil profile. In contrast, surface cracks are less impacted by root activity and have a limited effect on deeper soil hydrology. Focusing on the crack profile enables a more comprehensive understanding of root-soil interactions and their influence on soil hydraulic properties. RAR and CIF were obtained from minirhizotron images, capturing images from the observation box side windows. This non-destructive method estimates root and crack dynamics in the field without disturbing the soil structure. Before quantitative analysis, the captured images underwent a series of digital image processing steps (Cheng et al., 2021; Liu et al., 2013) to prepare them for detailed examination. These steps included grayscale conversion, which simplified the analysis by focusing on intensity variations, and binarization, which classified pixels as either black or white to distinguish between roots, cracks and soil. Denoising was then applied to enhance clarity by removing noise from the images. Skeletonization followed, reducing the structures to their basic forms to facilitate measurement. Finally, crack identification highlighted the cracks within the binary images. RAR can be expressed using the following equation:

$$RAR = \sum_{i=1}^{n}\frac{A_r}{A}x\,100\% \tag{6}$$

Where A_r is the number of root pixels and A is the total number of pixels. This ratio measures the proportion of soil area occupied by roots. CIF was assessed by dividing the crack area by the total area, as shown in the following equation:

$$CIF = \sum_{i=1}^{n}\frac{a_c}{A}x\,100\% \tag{7}$$

where a_c is the sum of all soil crack pixels, and A is the total number of pixels. It should be noted that a rectangular shape was adopted for the determination of RAR and CIF to address computational efficiency. This approach necessitated the exclusion of potential distortions attributable to the double ring side.

2.4. Test procedure

The experimental procedure began with the transplantation of vetiver tillers into the field at the beginning of October 2021. This timing was strategically chosen to coincide with the end of the rainy season, ensuring that the initial establishment period for the vetiver grass would benefit from the remaining high soil moisture conditions. Following the transplantation, the DRI were installed one week later. This interval allowed the vetiver tillers to begin acclimating to the soil environment, ensuring that their root systems could start interacting with the surrounding soil, thereby providing a more stable basis for subsequent measurements. The first infiltration test was conducted two weeks after the installation of the DRI setup. This period was sufficient for any initial soil disturbances caused by the installation to stabilise, ensuring that the infiltration measurements accurately reflected the influence of the vetiver roots on the soil structure. Additionally, the entire observation test was conducted over a period of 1.5 years, allowing for the observation of both wet and dry periods across two cycles. Given that Vetiver grass reaches maturity approximately 6 months after planting, this timeframe allowed for the investigation of two distinct developmental stages of the grass: the growth phase and the post-maturity decay phase within each cycle of wet and dry conditions. In Thailand, the climate is classified into three distinct periods: the rainy season (May to October), the dry winter (November to January), and the dry summer (February to April) (Meteorological Department of Thailand, 2020). The weather data such as air temperature, relative humidity, light and wind speed were measured using the weather station located in the test area. Furthermore, the climatic data was used to calculated Water Vapor Pressure Deficit (WVPD). This parameter measures the difference between the amount of water vapor that is actually present in the air and the maximum amount of water vapor the air can hold at a given temperature which plays a significant role in both evaporation and transpirations (Abtew and Melesse, 2013). WVPD was calculated based using Eqs. (8)–(10) (Dingman, 2015).

$$WVPD = e^{*} - e \tag{8}$$

$$e^{*} = 611.exp\left(\frac{17.27\,T}{T + 237.3}\right) \tag{9}$$

$$e = RH.e^{*} \tag{10}$$

Whereas e* is saturation vapor pressure (Pa), e is actual vapor pressure (Pa), T is air temperature (°C) and RH is relative humidity (%).

Throughout the 18-month monitoring period, the infiltration tests and image analyses provided comprehensive data on how seasonal variations influenced root growth, crack formation, and soil hydraulic properties. The collected data was analysed to determine the correlation between weather conditions, RAR, CIF, and K_{sat}. This detailed examination provided valuable insights into the dynamics of soil-vegetation interactions under varying seasonal conditions.

3. Results and discussion

3.1. Spatiotemporal analysis of crack distributions and crack intensity

Fig. 3 illustrates the variation in rainfall and WVPD during observation periods. These parameters provide insight into the relationship between precipitation and atmospheric drying power across different climatic periods (1st Rainy season, 1st dry Winter, 2nd Dry Summer, 2nd rainy season, 2nd dry Winter, and 2nd Dry Summer). During the rainy seasons (October 2021–January 2022 and June 2022–October 2022), higher levels of precipitation were observed. Rainy seasons show higher rainfall and generally lower WVPD values. This indicates a more humid atmosphere, as high precipitation events reduce the drying capacity of the air. In contrast, during the dry winter and dry summer, rainfall decreases significantly, with sporadic or negligible precipitation events shown by shorter blue bars or complete absence of bars in some cases. Dry seasons, particularly the Dry Summer, exhibit a contrasting pattern with almost no rainfall and an increase in WVPD, indicating a drier atmosphere.

Fig. 4 illustrates the spatiotemporal distributions of cracks in Zone B over an 18-month-long field study. The representative photos were taken from the observation hole marked by a white dotted rectangle in Fig. 1. The monochromatic photographs at the bottom display binarized representations of the cracks. Observations were made before each infiltration test. The CIF represents the crack intensity factor derived from the specific photo, while avgCIF represents the average CIF calculated from photos across all observation holes. Throughout the 18-month study, the geometry and pattern of cracks varied significantly with the seasons. Initial observations in October 2021, at the tail end of the rainy season, showed minimal crack formation with a crack diameter (CD) of 1.1 mm and a CIF of 0.9 %. The cracks were small and sparse, reflecting high soil moisture levels from the preceding months of rainfall. The increased soil moisture was a result of the substantial rainfall observed during this period, as indicated in Fig. 3, which recorded nearly 60 mm. The cracks were short and discontinuous, delineated by a red-dotted oval. The avgCIF was 0.8 %, indicating a low overall crack density. By January 2022, drier conditions had led to significant changes in the soil structure. The crack diameter increased to 2.7 mm, and the CIF rose to 3.2 %, with an avgCIF of 3.0 %. Cracks became more pronounced, with increased length and connectivity, forming more continuous and complex patterns. This period marked the beginning of more extensive and interconnected crack networks due to reduced soil moisture. The substantial increase in crack dimensions could be attributed to sustained evaporation, leading to soil desiccation and additional cracks, as elucidated by Piroird et al. (2016). In February 2022, dry winter conditions persisted might be as the result of the increasing of evaporation due to the higher WVPD that averagely reached 1.15 kPa (see Fig. 3), leading to further soil drying and contraction. WVPD significantly influences evaporation rates whereas higher WVPD creates a greater moisture difference between the air and surface, accelerating water movement from the surface to the air. (Abtew and Melesse, 2013). The crack diameter increased to 2.9 mm, with a CIF of 3.3 % and an avgCIF of 4.1 %. Cracks continued to expand in length and width, forming more intricate and interwoven patterns. The crack geometry showed a higher degree of connectivity, indicating progressive soil desiccation and structural changes. April 2022 marked the beginning of the dry summer season. The crack diameter peaked at 3.5 mm, with a CIF of 4.4 % and an avgCIF of 5.2 %. This crack become more prominent might be due to the influence of the increasing of WVPD which has reached 1.4 kPa. When temperature increased and relative humidity lower, they would result higher evaporation rate hence decrease the water content (Yang et al., 2023). Cracks became the widest and most extensive observed during the study, forming deep, continuous paths. The crack networks were highly interconnected, with complex geometries indicative of severe soil desiccation.

By June 2022, the onset of the rainy season had brought significant changes in soil cracking. The maximum crack diameter decreased to 1 mm, suggesting a slight recovery in soil moisture. The CIF was 3.2 %, with an avgCIF of 3.8 %. This decreasing of CIF might be due to the higher soil moisture content in this period as the result of rainfall during June 2022 reached up to 25 mm. Cracks remained significant but showed a slight reduction in width and length compared to April. The

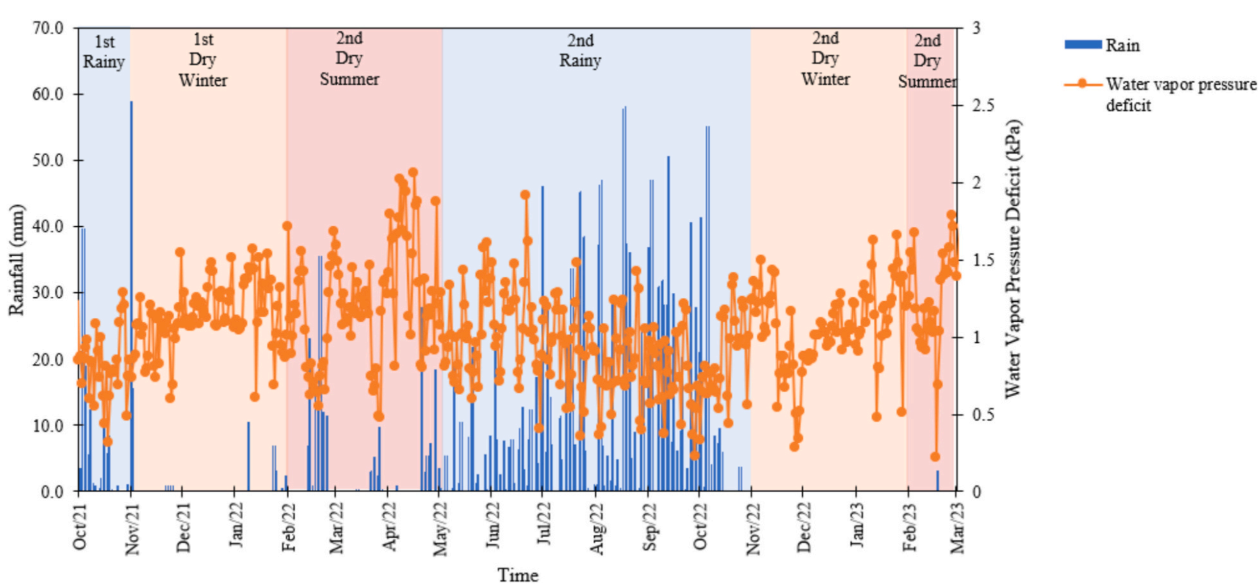

Fig. 3. The weather parameters during test period.

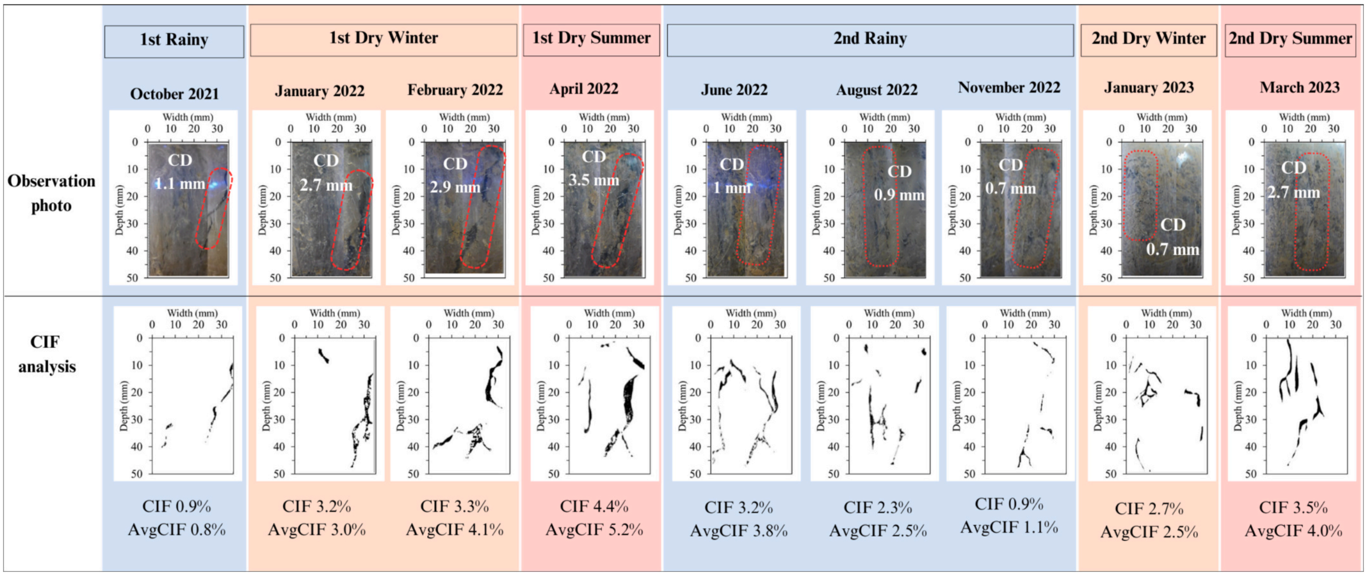

Fig. 4. Images showing temporal dynamics and analysis of desiccation cracks in the bare zone.

decreased crack diameter observed in August 2022 was 0.9 mm, slightly less than in June. The CIF was 2.3 %, with an avgCIF of 2.5 %. Cracks showed further reduction in width and length, becoming less interconnected, suggesting ongoing soil recovery from sustained rainfall. This reduction in crack width is likely the result of improved soil moisture conditions, possibly due to increased precipitation during these months. According to Tian et al. (2023), the closure of desiccation cracks in clayey soils is primarily linked to two factors: the swelling triggered by wetting and the breakdown of soil grain structure.

By November 2022, the crack diameter had decreased to 0.7 mm, suggesting the effects of preceding rainfall on desiccation. This decreasing of crack might be due to the substantial heavy rainfall occurred during July to October 2022 which reached up to 57 mm which almost similar from 1st rainy period as shown in Fig. 3. The CIF was 0.9 %, with an avgCIF of 1.1 %. In January 2023, the observed crack diameter remained at 0.7 mm, with an increase in CIF to 2.7 % and an avgCIF of 2.5 %. By March 2023, the crack diameter had increased to 2.7 mm. The CIF was 3.5 %, with an avgCIF of 4.0 %, reflecting increased crack density. The increase of this diameter agree with the study by Liaudat and Muraro (2024) found that the increase in CIF during drying stage was predominantly caused by the expansion of pre-existing cracks. From January 2023 to March 2023, more cracks appeared during the second drought period. As mentioned before, this crack was formed might be due to the increasing of WVPD that reached up to 1.7 kPa which can contribute to decrease of soil moisture due to evaporation process. On the other hand, the rainfall occurred in this period also minimize as shown in Fig. 3. these cracks were concentrated in a different location compared to the previous year, delineated by the red-dotted oval. This observation aligns with the findings by Liu et al. (2022), who similarly documented microcrack formation in virgin regions following drying-wetting cycles, mirroring the results obtained in this investigation. The remaining microcracks formed during the wetting process led to further expand in width and depth of the cracks in subsequent drying cycle. Over time, these cracks gradually evolved into a rough and irregular shape (Liu et al., 2024)The emergence of microcracks can be attributed to the roughening of surface edges from an elevated number of drying-wetting cycles (Tang et al., 2011). The repetition of wetting and drying sequences may compromise soil integrity, establishing zones of weakness within the soil matrix (Cheng et al., 2021; Yong and Warkentin, 1975). These weakened zones serve as focal points for the accentuated development of cracks, particularly during subsequent drying-wetting cycles (Tang et al., 2020).

3.2. *Seasonal dynamics of root growth and decay*

Fig. 5 presents a photo of Zone V2 taken from the observation hole within the planting row. The root structure is pixelated in the monochromatic photograph at the bottom of Fig. 4. The RAR indicates the ratio between the root area and the total area of the image, while avgRAR represents the average RAR calculated from root observation images obtained from all observation sites. During the initial observation in October 2021, approximately one week after grass planting, the roots were still in the process of establishing themselves within the area. This phase represents a typical stage in plant ontogeny, characterised by initial root dispersal within the substrate, serving to stabilise the plant and facilitate the procurement of water and nutrients (Sorensen, 2000). Given the relatively short duration since planting, substantial root proliferation is improbable, thereby corroborating the observed phase of root system establishment.

By January 2022, three months after planting, the identification of fine roots in the image aligns with the expected sequential development stages of root systems in plants. Initially, plants prioritise the growth of fine roots, which are essential for the efficient absorption of water and nutrients. As the plant progresses in its growth cycle, the root system undergoes expansion, marked by the emergence of coarse roots, leading to an increase in avgRAR to 1.2 %. This latter phase of development, evident from February 2022, is highlighted by the presence of coarse roots with a root diameter (RD) reaching approximately 2.3 mm (delineated by the yellow dotted oval) and avgRAR reaching up to 2.2 %. Such roots are instrumental in providing structural support and indicate the plant's adaptive response to the soil's physical and nutritional environment (Ghestem et al., 2011; Liu et al., 2009).

The photographic evidence, capturing the transition from fine to more substantial coarse roots, reflects the typical dynamic adaptation of plant root systems in response to environmental variations. Throughout this growth period, the roots continued to extend significantly, reaching an RD of 2.7 mm and an avgRAR of 4 % by April 2022. This higher root density and deeper root systems can enhance a plant's ability to access water from a larger soil volume as the result of higher WVPD reached up to 2 kPa (Fig. 3) which can lead to increased evapotranspiration. WVPD is also one major factor that can influence the evapotranspiration rate (Abtew and Melesse, 2013; Elbeltagi et al., 2023). High VPD creates a significant difference between the water vapor content inside the plant and the atmosphere, which can lead to increased water loss through transpiration (Devi and Reddy, 2018; Grossiord et al., 2020). During the

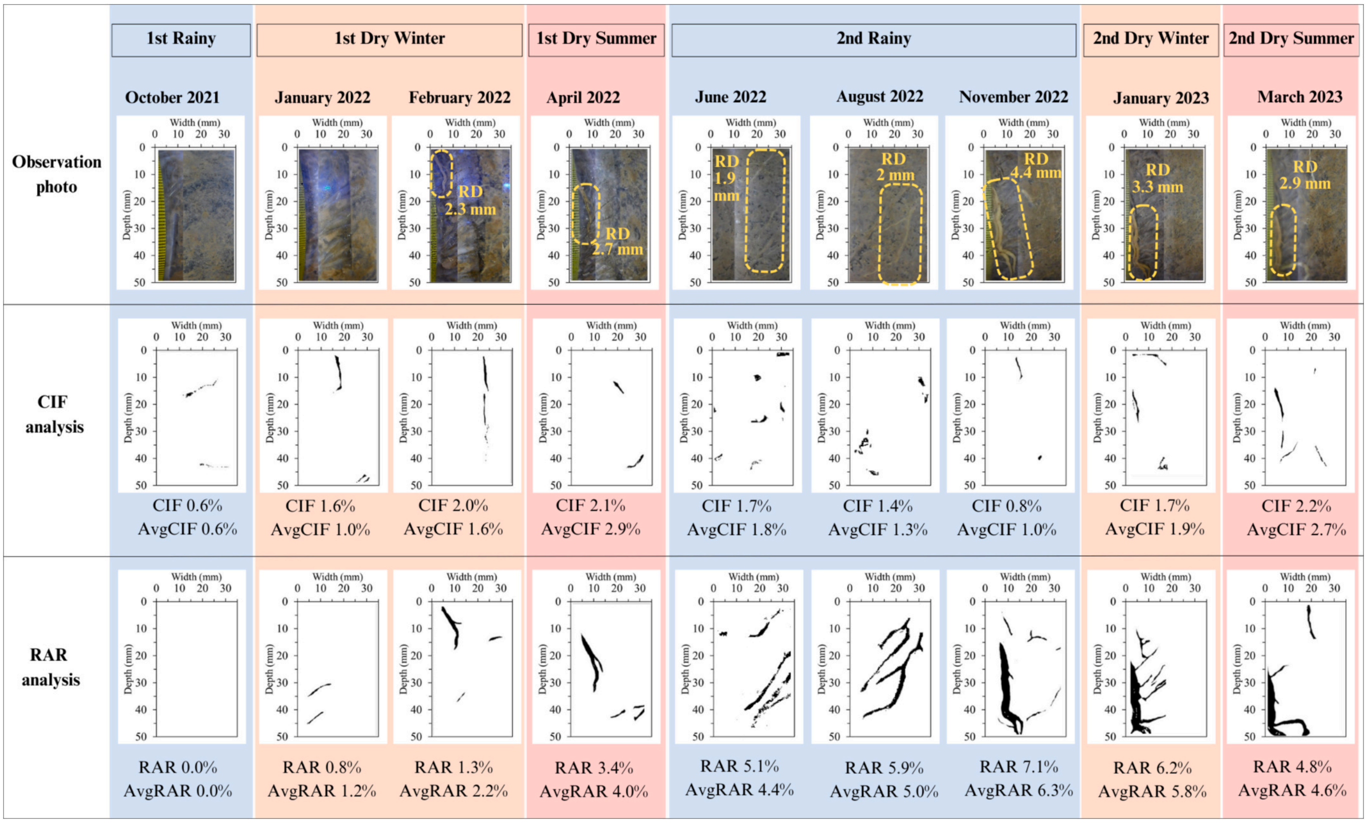

Fig. 5. Images showing temporal dynamics and analysis of desiccation cracks and roots in the zone V2.

summer drought period, the roots were not clearly captured in the selected image, although the average value of RAR gradually increased due to the presence of relatively fine roots emerged in other areas. It is conceivable that the coarse roots extended deeper into the soil during this dry summer to access additional moisture from the lower layers (Dawson, 1993). In November 2022, roots in the same area observed in April 2022 were clearly captured. These roots exhibited more tortuosity, with an RD of 4.4 mm (1.6 times larger), confirming that the root was still in the active growth phase, as indicated by the increase in avgRAR to 6.3 % during the first year. This observation corroborates that the moisture near the surface as the result of the heavy rainfall (up to 57 mm) occurred during this 2nd rainy period was sufficient for the roots, obviating the need for them to extend deeper. This behaviour highlights the adaptability of roots, which can adjust both in size and location in response to prevailing climatic conditions.

Consequently, the RD decreased by 1.3 times (to RD = 3.3 mm) in January 2023 during the dry winter, as delineated by the yellow dotted oval, resulting in a reduced avgRAR of 5.8 %. The RD continued to decrease to 2.9 mm and the avgRAR decreased to 4.6 % by March 2023. This phenomenon suggests that possible dehydration and root decay during this period likely contributed to the observed root shrinkage. As shown in Fig. 3, the WVPD during this time reached to 1.7 kPa and no rainfall occurred during this period. This condition contributed to the high transpiration which leading to decrease of soil moisture content. As the result, plants might experience water stress and dehydration. Root dehydration could lead to reduced turgor pressure and shrinkage. Similarly, potential root decay, possibly due to terminal water stress and root turnover, might further deteriorate the root structure (Kamchoom et al., 2023; Kamchoom et al., 2022b; Richardson et al., 2009).

3.3. *Influence of seasonal root dynamics on soil crack formation*

When observing cracks in Zone V2, the pattern appears less visible compared to the distribution of cracks in Zone B. Unlike the uniformly distributed cracks in Zone B, the crack distribution in the vetiver zone was generally uneven, likely due to the presence of vetiver roots. The findings of Cheng et al. (2023) corroborate this observation, revealing that vegetated soil exhibits smaller values in both CIF and average crack width compared to their counterparts without vegetation. Moreover, an increase in planting density appears to be associated with a higher frequency of finer and thinner cracks. In addition to enhancing soil tensile strength through root reinforcement, root intrusion in macropores can reduce the volume of desiccation cracks (Bordoloi et al., 2020). The plant roots thus occupy a portion of the soil pores, limiting the drying-induced soil shrinkage and leading to a reduction in the final degree of cracking.

Initial observations in the vetiver zone indicated the presence of only minor cracks. However, thin cracks with an approximate length of 15 mm started to appear in January 2022, resulting in an avgCIF of 1.0 %. Subsequently, during the drought period extending up to April 2022, these cracks exhibited a gradual expansion, eventually attaining a maximum width of approximately 0.8 mm and a length of around 35 mm in February 2022. Moreover, this crack expansion led to an increase in avgCIF to 2.9 % in April 2022. During drought periods, soil moisture decreases due to higher WVPD (reached up to 2 kPa). As the soil shrinks, the roots may not be able to counteract the forces causing the cracks to widen compared to previous period (1st rainy period). It should be noted that cracks in vegetated soil were four times less wide than those observed in Zone B during the same period. During the wetting period in November 2022, a significant reduction in the prevalence of cracks was observed, resulting in an avgCIF of 1 %, similar to the avgCIF in Zone B. Notably, the cracks that persisted were situated away from the root structures, as depicted in Fig. 5. This observation suggests that the wetting process played a role in mitigating the extent of cracks, particularly those distant from the root zones.

During the period from January to March 2023, there was a lack of rainfall and a significant increase in WVPD, reaching a peak of 1.7 kPa. This condition made the crack in zone V2 expanded 2.7 times (AvgCIF become 2.7 %) compared to 2nd rainy period. More importantly, during this period observations revealed instances of root decay, with avgRAR exhibiting a further decrease to 4.0 %. Consequently, during this second drought period, cracks reappeared resulted almost two times higher avgCIF compares with previous rainy period, primarily manifesting at the interface between the soil and roots. However, this increase of AvgCIF during 2nd dry period at zone V2 still was 1.7 times lower compared to zone B. While it may be deduced that cracks existing before or during root development could be minimised due to root propagation within the area, it is noteworthy that cracks may reappear more readily at the soil-root interface during periods of root decay. This observation underscores the complex interplay between root dynamics and soil cracking processes, highlighting the need for further investigation into the mechanisms underlying crack formation and propagation under the influence of vegetation.

3.4. K_{sat} variations under seasonal changes

Fig. 6 provides a comprehensive overview of the temporal variations in K_{sat} across different seasons, highlighting the differences between Zone B and the V1 and V2 zones. The K_{sat} values are plotted over an 18-month period, encompassing two wet seasons, two dry winter seasons, and two dry summer seasons. Initially, the K_{sat} value was approximately 2.9×10^{-7} m/s. Following exposure to dry conditions until the end of winter, there was a progressive increase in K_{sat} values across all zones, notably in Zone V2. This increase in K_{sat} persisted in Zone V2 throughout the dry conditions, reaching approximately 4.5×10^{-6} m/s, marking a significant 16-fold increase compared to the initial value. This increment in K_{sat} appeared consistent throughout the entire dry period, encompassing both the winter and summer seasons. These findings align with prior research by Leung et al. (2018), highlighting a positive correlation between below-ground traits such as root biomass and root length density with K_{sat}. In this study, the increase in K_{sat} by about 16 times corresponded to an average rise in root biomass equal to 4 % at six months of plant age. This observation underscored the potentially pivotal role of root propagation during the establishment phase. As roots expand, they often create more micropores (Juyal et al., 2021) allowing for faster and more efficient water transport compared to the surrounding soil matrix.

In Zone V1, a more modest increase in K_{sat} was observed, resembling the behaviour in Zone B. This raises the possibility of substantial variability in root development between Zones V1 and V2 during the initial stages of the study. However, during the dry summer, K_{sat} in Zone V1 experienced a sudden rise to 3.4×10^{-6} m/s, thereby approaching the value observed in Zone V2. This fluctuation is likely attributed to root

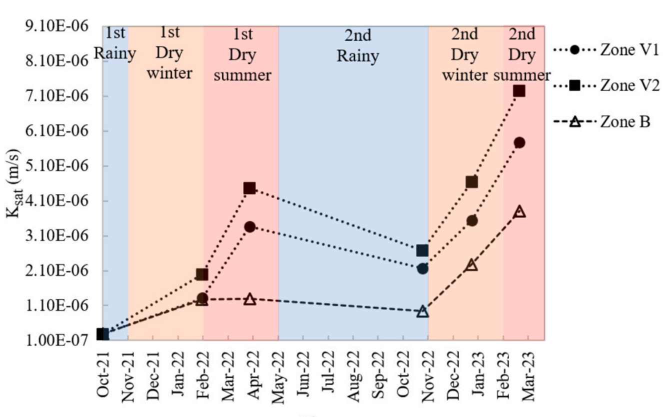

Fig. 6. Seasonal variability of saturated hydraulic conductivity (K_{sat}) in bare and vegetated zones.

development in Zone V1, which may have initially been restricted but gradually aligned with that of Zone V2, subsequently exerting an impact on the increase in K_{sat}. In contrast, the K_{sat} within Zone B demonstrated a marginal increase throughout the summer period, substantially less in magnitude compared to that in the vegetated zones. Remarkably, this relative constancy in K_{sat} was maintained even amidst an elevation in the CIF from 4.1 % to 5.2 %, a figure notably double that observed within the vegetated zones. In fact, the observed increase in CIF within Zone B, attributed primarily to the expansion of pre-existing cracks (as indicated in Fig. 3), seems to have had a negligible effect on the K_{sat} values. The findings suggest that root development may mitigate the increase in CIF during the summer, thereby resulting in the reduction in K_{sat}. However, it also establishes preferential flow paths during its propagation. It appears that these preferential flow path supersede the effects of soil crack reduction and lead to an overall increase in K_{sat}. The presence of cracks induces a more pronounced soil deformation pattern, characterised by both swelling and subsidence (Sadeghi et al., 2024).

During the transition from the first summer to the second rainy period, K_{sat} values in different zones exhibited varying patterns. In Zone V2, K_{sat} significantly reduced from 4.5×10^{-6} m/s to approximately 2.7×10^{-6} m/s. In contrast, Zone V1 showed a reduction of K_{sat} from 3.4×10^{-6} m/s to 2.1×10^{-6} m/s, and Zone B exhibited a slight reduction from 1.3×10^{-6} m/s to approximately 9.5×10^{-7} m/s. The slight reduction in K_{sat} in Zone B could be attributed to soil swelling during wetting. However, the observed changes in K_{sat} in Zone V2, which are more significant than those attributable to mere soil crack closure from wetting where Pearson correlation between CIF and K_{sat} see reached up to 0.835 (see Table 2), can be ascribed to a synergistic effect of soil swelling and extensive root-system development. The well-established root system likely induced notable alterations in soil structure, such as altered pore distribution. These changes could account for the pronounced decrease in K_{sat}. Contrastingly, Zone V1, also subject to the effects of soil swelling and root development, exhibited a less marked change in K_{sat}. This may suggest a lower degree of root development, leading to comparatively milder modifications in soil structure and hydraulic properties than in Zone V2.

In the transition from the second rainy season to the second summer, a consistent increase in K_{sat} values was observed across both Zone V1 and Zone V2, mirroring the trend noted in the initial year. However, the K_{sat} values during this period were significantly higher than those recorded in the first year. Specifically, Zone V1 exhibited a K_{sat} of approximately 5.8×10^{-6} m/s, while Zone V2 showed a K_{sat} of around 7.2×10^{-6} m/s, indicating a 1.6-fold increase from the first summer's values. The observed increase in K_{sat} during the second summer period, with higher values in both zones V1 and V2, can be linked to the decay of plant roots and the consequent increase in soil porosity. This period is characterised by a decrease in RAR, indicative of root decay, contrasting with the initial year, where an increase in RAR suggested ongoing root growth. The findings are consistent with Rajamanthri et al. (2021), who noted a positive correlation between K_{sat} and plant age in vetiver-planted soil. They projected an increase in K_{sat} to 5×10^{-5} m/s at around 20 months of plant age, during which time some plant roots were observed to be decaying.

It is important to note that the earlier observed variations in K_{sat} between Zones V1 and V2, attributed to natural differences in root development, were no longer evident in the decay phase during first year observation. The K_{sat} values in both zones increased similarly, suggesting the interconnected formation of preferential flow paths due to root decay had a significant impact on K_{sat}. This indicates that the effects of decaying roots on soil hydraulic conductivity are significant and consistent, overshadowing the variations in root amounts due to natural differences in root development. Furthermore, Zone B exhibited a notable increase in K_{sat}, registering a value of 3.8×10^{-6} m/s. This value marked a substantial 2.9-fold rise compared to the K_{sat} observed in the initial summer. Louati et al. (2021) also reported an increase in permeability with the number of wetting-drying cycles, particularly notable in clayey soils without vegetation. This observation gains significance given that the recently compacted topsoil in this study, situated approximately half a metre deep, implies a potential for more pronounced effects from successive wetting and drying cycles before reaching residual conditions. The significant increase in K_{sat} observed during this period may be attributed to the complex network of interconnected cracks, as evidenced in Fig. 3 in March 2023 compared to April 2022. This network facilitates enhanced water flow, as discussed in studies by Cheng et al. (2021) and Louati et al. (2021). Consequently, water can rapidly traverse these pathways, leading to a higher hydraulic conductivity despite the similar average CIF value.

The increase in K_{sat} in Zone B during the second dry period mirrored the rise observed in Zone V2 during the same timeframe, indicating an influencing soil hydraulic conductivity. In Zone V2, this increase is attributed to the enhancement of preferential flow pathways, formed by the decay of roots. In contrast, in Zone B, similar pathways arise due to soil cracking. These preferential pathways, whether from decaying roots or soil cracks, increase macro-porosity and enhance water flow during dry periods. The decay of roots in vegetated soils, therefore, plays a crucial role in influencing soil hydraulic conductivity, akin to physical soil changes in Zone B. Moreover, the higher K_{sat} values in vegetated zones may be attributed to the presence of residual 4.6 % RAR that could continue to contribute to decay processes and further enhance soil porosity. This increased porosity, resulting from both decaying roots and potential soil cracks, creates additional pathways for water flow, ultimately leading to higher K_{sat} values in Zone V2 compared to Zone B.

3.5. *Correlation between CIF, RAR and hydraulic conductivity ratio (K_{sat} ratio)*

Fig. 7 illustrates the correlation between CIF and K_{sat} ratio within Zone B. The K_{sat} ratio was calculated by dividing the K_{sat} value at a given time by the initial K_{sat} value, which was approximately 2.9×10^{-7} m/s. This significant relationship underlines the pivotal role of cracks in shaping soil hydraulic conductivity. As shown in Table 2, K_{sat} was highly influenced by CIF with respect to Pearson corelation of 0.83506. The figure clearly indicates that drying and the summer period result in the CIF increase path, leading to an increase in the K_{sat} ratio. On the other hand, the wetting period resulted in a decrease the CIF path, which influenced the decrease in the K_{sat} ratio. This can be proven by Pearson corelation of K_{sat} and CIF to rainfall (wetting period) was −0.5658 and − 0.65229, respectively. However, it is noteworthy that the decreasing trend line did not align perfectly with the increasing trend, indicating distinct hysteresis between CIF and the K_{sat} ratio.

During rainy periods, soil swelling causes finer cracks to close, leading to a decrease in CIF. However, larger cracks, especially those formed due to desiccation processes, tend to remain open even after wetting events, serving as pathways for enhanced water flow. Interestingly, during the subsequent dry winter and summer period, CIF increased up to 4 % in March 2023, which was almost the same as in February 2022 (CIF of 4.1 %). However, the K_{sat} ratio exhibited a significant rise to 12, almost three times higher than the K_{sat} ratio in February 2022. The increase in CIF during dry periods is typically accompanied by the formation of new and finer cracks, which enhance soil permeability and result in a higher K_{sat} ratio. Conversely, during wet periods, soil swelling, and the closure of larger cracks reduce CIF, leading to a lower K_{sat} ratio. This complex interaction between crack formation, crack closure, and soil hydraulic conductivity indicates that the soil's response to drying and wetting cycles is not linear. Instead, it demonstrates hysteresis, where the history of crack development and closure influences the current state of K_{sat}. Understanding these pathways is crucial for accurately predicting soil hydraulic behaviour under varying environmental conditions and for developing effective soil management and erosion control strategies.

Fig. 8 illustrates the intricate relationship between RAR, CIF, and K_{sat} ratio within Zone V2. This graph provides compelling evidence of the varied impacts of different stages of root growth on soil hydraulic conductivity, showcasing both augmentative and inhibitory effects. The preference for selecting zone V2 in this discussion arises primarily from the observation of root, which was exclusively installed in the zone V2 as depicted in Fig. 2. As a result, V2 provides complete data regarding both root growth and decay. This configuration facilitates a direct correlation between root growth, decay, and K_{sat}. During the RAR increase path, root propagation potentially plays a pivotal role in increasing the K_{sat} ratio, as discussed in the previous section. The RAR gradually increased to 4 %, resulting in an increase in the K_{sat} ratio up to 15.6 in April 2022 whereas the Pearson correlation analysis revealed a coefficient of 0.62496 between K_{sat} and RAR. Following this, the RAR experienced a gradual increase to 6.3 % due to an increase in root diameter rather than active propagation during the winter period (as in Fig. 4 in November 2022). As the root diameter enlarges, the pre-existing network of channels and pores within the root zone contracts. This, combined with the preventive effects on nearby cracks by the roots and crack closure from wetting, leads to a subtle reduction in the K_{sat} ratio to about 9.34.

Following over a year of observation, the roots initiated a decaying process, as detailed in the previous section. This decay led to a discernible reduction in the RAR. However, it is noteworthy that this decrease in RAR did not correlate with a reduction in the K_{sat} ratio. During the final observation conducted in March 2023, the RAR continued its decline, reaching 4.6 %, which led to an even higher K_{sat} ratio of 25. This observation indicates that, despite the reduction in root density, the soil's hydraulic conductivity experienced a notable increase. As the root system decayed further, the root channels became more prominent, enhancing the pathways for water infiltration within the soil

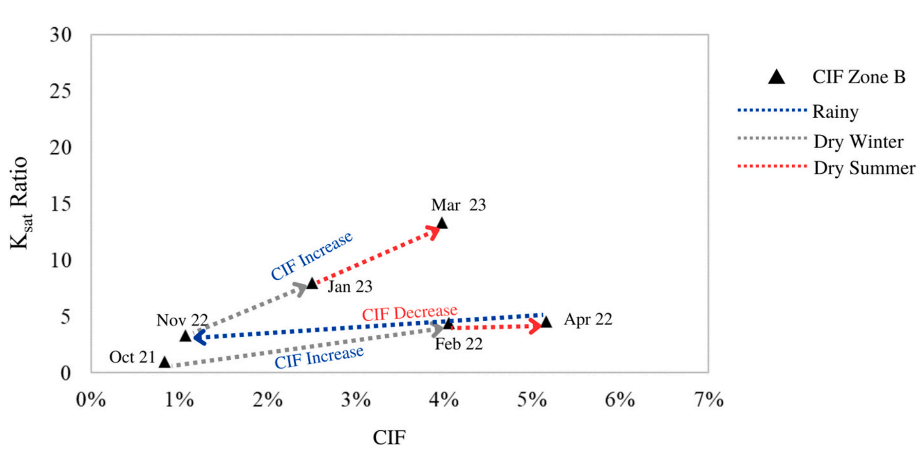

Fig. 7. Hysteretic correlation between crack intensity factor (CIF) and saturated hydraulic conductivity (K_{sat}) ratio in the bare area.

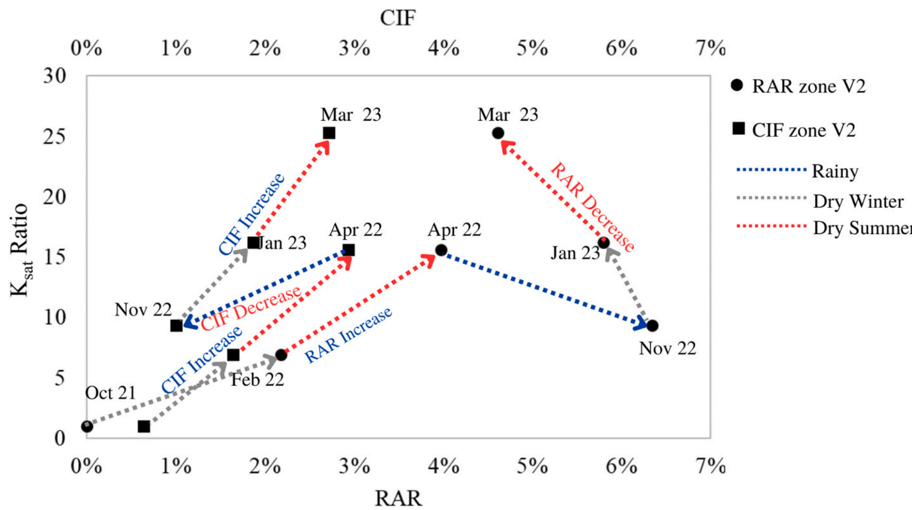

Fig. 8. Hysteretic correlation between root area ratio (RAR), crack intensity factor (CIF) and saturated hydraulic conductivity (K_{sat}) ratio in the zone V2.

(Guo et al., 2019; Wu et al., 2021; Zhang et al., 2017).

This finding underscores a crucial insight: the hydraulic conductivity of vegetated soil does not follow a linear relationship with root biomass and soil cracks. Instead, it exhibits hysteresis behaviour. For instance, a root biomass of 4.5 % may have lower cracks and hydraulic conductivity if it is on an increasing pathway (i.e., during a growth stage or when it possessed a lower root biomass previously). Conversely, the same root biomass of 4.5 % may exhibit more cracks and higher hydraulic conductivity if it is on a decreasing pathway (i.e., during a decay stage or when it had a higher root biomass previously). This challenges the conventional concept from prior literature linking only root biomass to soil hydraulic conductivity, emphasising the necessity of considering the growth and decay pathways of the root system.

Furthermore, a comparative analysis of the K_{sat} ratio increase between zones V2 and B reveals that the increase within the bare area showed a more pronounced trend, especially in the second year. This trend was noticeable from November 2022 to March 2023, although the magnitude remained lower than that observed in the vegetated zone. Conversely, a more subdued increase was observed in the vegetated area, even during periods of root decay, as illustrated in Fig. 8. This difference highlights the faster rise in K_{sat} in bare soil, which could lead to potential challenges ahead. Consequently, these findings indicate a more stable rise in hydraulic conductivity for the vegetated areas, demonstrating their potential to mitigate the adverse impacts of climate variations on soil hydraulic conductivity. This phenomenon holds promise for enhancing the overall adaptation of earthworks to changing and more extreme climatic conditions.

Although this study considers Vetiver grass which has a high root density, deep penetration, and robust biomass production, exploring the influence of other plant species with varying root architectures and biomass are critical for broadening their applicability in diverse environmental and geotechnical contexts. Fibrous-rooted plants, such as Bermuda grasses, typically exhibit shallow root systems (Xu et al., 2023) that influence near-surface cracks during primary growth stages. This can enhance rapid infiltration but limit deeper water movement, resulting in a distinct K_{sat} profile. In contrast, taproot-dominated species, such as trees, are characterised by deep, singular roots with minimal lateral spread (Stokes et al., 2008). These roots play a critical role in the continuity of subsurface cracks and facilitate deeper water flow pathways. Such dynamics in root geometry could have important implications for groundwater recharge (Ghestem et al., 2011; Guo et al., 2019) and slope stability (Leung et al., 2017; Ng et al., 2016). Additionally, species with high root turnover rates, such as annual crops, exhibit frequent root decay (Wei et al., 2019), leading to transient changes in soil structure. These changes could create temporary preferential flow paths or alter the continuity of cracks. Understanding these temporal dynamics in root biomass is essential for assessing soil-water interactions and hydrological processes.

4. Conclusion

This research investigates the spatiotemporal variations in crack formation, root dynamics, and their combined influence on K_{sat} under diverse climatic conditions. The findings indicate that the geometry and pattern of desiccation cracks vary significantly with the seasons, showing minimal formation during the rainy season and increased crack width, length, and connectivity during dry periods, peaking at a CIF of approximately 5.2 %. Spatio-temporal changes in RAR for vegetated soil highlight initial root establishment, subsequent growth phases marked by increased root diameter and coarse root emergence with RAR rising to 6.3 %, and root shrinkage due to dehydration and decay, dropping the RAR to 4.6 %. The presence of vetiver roots results in less visible and uneven crack distribution compared to bare soil, with smaller CIF by half and average crack widths. The reappearance of cracks during root decay periods underscores the complex interplay between root dynamics and soil cracking processes.

The findings confirmed K_{sat} values increased significantly during dry periods, with the most substantial rise attributed to root propagation, reaching a 16-fold increase in the vegetated zone, whereas the bare zone showed only a marginal 5-fold increase. This suggests that cracks alone had a lesser impact compared to the combined effects of root development and soil cracking in vegetated zones. During the transition from the dry summer to the rainy season, K_{sat} decreased due to soil swelling with the vegetated zone experiencing the most notable reduction. However, during the second dry period, K_{sat} values in the vegetated zone increased again, attributed to root decay and the formation of preferential flow paths. The findings also demonstrate that K_{sat} is significantly influenced by the presence of cracks and roots, exhibiting distinct hysteresis behaviour in response to drying and wetting cycles. A root biomass of 4.5 % can result in lower cracks and hydraulic conductivity during growth phases but exhibit more cracks and higher hydraulic conductivity during decay phases. This finding has significant implications for engineering geology design, particularly in projects involving slope stability, landfill cover design, and erosion control. The observation that the K_{sat} of vegetated soil exhibits hysteresis behaviour, rather than a simple linear relationship with root biomass and soil cracks, challenges traditional models and assumptions. This insight emphasises the importance of accounting for the dynamic nature of root systems, including their growth and decay pathways, when designing geotechnical solutions.

CRediT authorship contribution statement

Yuliana Yuliana: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Arwan Apriyono:** Writing – review & editing, Validation, Methodology, Investigation, Data curation. **Viroon Kamchoom:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition, Formal analysis, Conceptualization. **David Boldrin:** Writing – review & editing, Validation, Investigation. **Qing Cheng:** Writing – review & editing, Validation, Investigation. **Chao-Sheng Tang:** Writing – review & editing, Validation, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.enggeo.2025.107973.

Data availability

Data will be made available on request.

References

Abtew, W., Melesse, A., 2013. Evaporation and Evapotranspiration. Springer, New York.

Apriyono, A., Yuliana Kamchoom, V., 2023. Serviceability of cut slope and embankment under seasonal climate variations. Acta Geophys. https://doi.org/10.1007/s11600-022-00953-x.

ASTM D 3385, 2003. Infiltration rate of soils in field using double-ring infiltrometer. Test 04, 4–11.

ASTM D1556, 2007. Standard test method for density and unit weight of soil in place by sand-cone method. D1556–07.

ASTM D3385-09, 2016. Standard test method for infiltration rate of soils in field using double-ring infiltrometer. In: ASTM Standards International. West Conshohocken, p. 7.

ASTM D4318, 2014. Standard Test Methods for Liquid Limit, Plastic Limit, and Plasticity Index of Soils, D4318–10. ASTM international West Conshohocken, PA.

Bagarello, V., Iovino, M., Elrick, D., 2004. A simplified falling-head technique for rapid determination of field-saturated hydraulic conductivity. Soil Sci. Soc. Am. J. 68, 66–73. https://doi.org/10.2136/sssaj2004.6600.

Boldrin, D., Knappett, J.A., Leung, A.K., Brown, J.L., Loades, K.W., Bengough, A.G., 2022. Modifying soil properties with herbaceous plants for natural flood risk-reduction. Ecol. Eng. 180, 106668. https://doi.org/10.1016/j.ecoleng.2022.106668.

Bordoloi, S., Ng, C.W.W., 2020. The effects of vegetation traits and their stability functions in bio-engineered slopes: a perspective review. Eng. Geol. 275, 105742. https://doi.org/10.1016/j.enggeo.2020.105742.

Bordoloi, S., Ni, J., Ng, C.W.W., 2020. Soil desiccation cracking and its characterization in vegetated soil: a perspective review. Sci. Total Environ. 729, 138760. https://doi.org/10.1016/j.scitotenv.2020.138760.

Chen, X.W., Kamchoom, V., Wu, J., Sun, G., Zhang, Q., Li, Huishan, Wong, M.H., Li, Hui, Wang, J., 2023. Potentials of mycorrhizal fungi in altering eight biomechanical properties of plant roots. L. Degrad. Dev. https://doi.org/10.1002/ldr.4955.

Cheng, Q., Tang, C.S., Xu, D., Zeng, H., Shi, B., 2021. Water infiltration in a cracked soil considering effect of drying-wetting cycles. J. Hydrol. 593, 125640. https://doi.org/10.1016/j.jhydrol.2020.125640.

Cheng, Q., Gu, Y.-D., Tang, C.-S., Zhang, X.-Y., Shi, B., 2023. Desiccation cracking behaviour of a vegetated soil incorporating planting density. Can. Geotech. J. 1–33. https://doi.org/10.1139/cgj-2023-0047.

Dawson, T.E., 1993. Hydraulic lift and water use by plants: implications for water balance, performance and plant-plant interactions. Oecologia 95, 565–574. https://doi.org/10.1007/BF00317442.

Devi, M.J., Reddy, V.R., 2018. Transpiration response of cotton to vapor pressure deficit and its relationship with stomatal traits. Front. Plant Sci. 871, 1–12. https://doi.org/10.3389/fpls.2018.01572.

Dingman, S.L., 2015. Physical Hydrology, 3rd, Editio. ed. Waveland Press, Inc., United State of America.

Elbeltagi, A., Srivastava, A., Deng, J., Li, Z., Raza, A., Khadke, L., Yu, Z., El-Rawy, M., 2023. Forecasting vapor pressure deficit for agricultural water management using machine learning in semi-arid environments. Agric. Water Manag. 283, 108302. https://doi.org/10.1016/j.agwat.2023.108302.

Elrick, D.E., Parkin, G.W., Reynolds, W.D., Fallow, D.J., 1995. Analysis of early-time and steady state single-ring infiltration under falling head conditions. Water Resour. Res. 31, 1883–1893. https://doi.org/10.1029/95WR01139.

Fatehnia, M., Tawfiq, K., Ye, M., 2016. Estimation of saturated hydraulic conductivity from double-ring infiltrometer measurements. Eur. J. Soil Sci. 67, 135–147. https://doi.org/10.1111/ejss.12322.

Gadi, V.K., Bordoloi, S., Garg, A., Sahoo, L., Berretta, C., Sekharan, S., 2017. Effect of shoot parameters on cracking in vegetated soil. Environ. Geotech. 5, 123–130. https://doi.org/10.1680/jenge.17.00013.

Gao, Q.F., Zeng, L., Shi, Z.N., 2021. Effects of desiccation cracks and vegetation on the shallow stability of a red clay cut slope under rainfall infiltration. Comput. Geotech. 140, 104436. https://doi.org/10.1016/j.compgeo.2021.104436.

Ghestem, M., Sidle, R.C., Stokes, A., 2011. The influence of plant root systems on subsurface flow: implications for slope stability. Bioscience 61, 869–879. https://doi.org/10.1525/bio.2011.61.11.6.

Gholami, M., Sadeghi, H., Alipanahi, P., 2024. Anisotropic hydraulic conductivity of as-compacted, bare and vegetated soils. Geotechnique. https://doi.org/10.1680/jgeot.23.00248.

Gregory, J.H., Dukes, M.D., Miller, G.L., Jones, P.H., 2005. Analysis of double-ring infiltration techniques and development of a simple automatic water delivery system. Appl. Turfgrass Sci. 2, 1–7. https://doi.org/10.1094/ats-2005-0531-01-mg.

Grossiord, C., Buckley, T.N., Cernusak, L.A., Novick, K.A., Poulter, B., Siegwolf, R.T.W., Sperry, J.S., McDowell, N.G., 2020. Plant responses to rising vapor pressure deficit. New Phytol. 226, 1550–1566. https://doi.org/10.1111/nph.16485.

Guo, L., Liu, Y., Wu, G.L., Huang, Z., Cui, Z., Cheng, Z., Zhang, R.Q., Tian, F.P., He, H., 2019. Preferential water flow: Influence of alfalfa (*Medicago sativa* L.) decayed root channels on soil water infiltration. J. Hydrol. 578, 1–8. https://doi.org/10.1016/j.jhydrol.2019.124019.

Huat, B.B.K., Ali, F.H.J., Low, T.H., 2006. Water infiltration characteristics of unsaturated soil slope and its effect on suction and stability. Geotech. Geol. Eng. 24, 1293–1306.

Jeswin, C., Marimuthu, B., Chithra, K., 2017. Ultrasonic water level indicator and controller using AVR microcontroller. In: 2017 Int. Conf. Inf. Commun. Embed. Syst. ICICES. https://doi.org/10.1109/ICICES.2017.8070773.

Juyal, A., Guber, A., Oerther, M., Quigley, M., Kravchenko, A., 2021. Pore architecture and particulate organic matter in soils under monoculture switchgrass and restored prairie in contrasting topography. Sci. Rep. 11, 1–13. https://doi.org/10.1038/s41598-021-01533-7.

Kamchoom, V., Leung, A.K., 2018. Hydro-mechanical reinforcements of live poles to slope stability. Soils Found. 58, 1423–1434.

Kamchoom, V., Leung, A.K., Ng, C.W.W., 2014. Effects of root geometry and transpiration on pull-out resistance. Geotech. Lett. 4, 330–336. https://doi.org/10.1680/geolett.14.00086.

Kamchoom, V., Boldrin, D., Leung, A.K., Sookkrajang, C., Likitlersuang, S., 2022a. Biomechanical properties of the growing and decaying roots of *Cynodon dactylon*. Plant Soil 471, 193–210. https://doi.org/10.1007/s11104-021-05207-1.

Kamchoom, V., Leung, A.K., Boldrin, D., Sakolpanya, T., Wu, Z., Likitlersuang, S., 2022b. Shearing behaviour of vegetated soils with growing and decaying roots. Can. Geotech. J. 59, 2067–2084.

Kamchoom, V., Chen, X.W., Leung, A.K., Sakolpanya, T., Srinil, C., 2023. Dynamic changes in cellulose content and biomechanical properties of mycorrhizal roots during growth and decay. Plant Soil 1–17. https://doi.org/10.1007/s11104-023-06103-6.

Lai, J., Ren, L., 2007. Assessing the size dependency of measured hydraulic conductivity using double-ring infiltrometers and numerical simulation. Soil Sci. Soc. Am. J. 71, 1667–1675. https://doi.org/10.2136/sssaj2006.0227.

Leung, A.K., Kamchoom, V., Ng, C.W.W., 2017. Influences of root-induced soil suction and root geometry on slope stability: a centrifuge study. Can. Geotech. J. 54, 291–303.

Land Development Department, 2014. Vetiver grass: plant for soil and water conservation [WWW Document]. URL. http://www1.ldd.go.th/ldd_en/en-US/vetiver-grass-plant-for-soil-and-water-conservation/ (accessed 11.17.23).

Leung, A.K., Boldrin, D., Liang, T., Wu, Z.Y., Kamchoom, V., Bengough, A.G., 2018. Plant age effects on soil infiltration rate during early plant establishment. Geotechnique 68, 646–652. https://doi.org/10.1680/jgeot.17.T.037.

Li, M., Liu, T., Duan, L., Luo, Y., Ma, L., Zhang, J., Zhou, Y., Chen, Z., 2019. The scale effect of double-ring infiltration and soil infiltration zoning in a semi-arid steppe. Water (Switzerland) 11. https://doi.org/10.3390/w11071457.

Liaudat, J., Muraro, S., 2024. Drying of silty soil treated with superabsorbent hydrogels: Retention behaviour and cracking, p. 331.

Liu, T.Y., Chang, C.Y., Chiou, T.J., 2009. The long-distance signaling of mineral macronutrients. Curr. Opin. Plant Biol. 12, 312–319. https://doi.org/10.1016/j.pbi.2009.04.004.

Liu, C., Tang, C.S., Shi, B., Suo, W. Bin, 2013. Automatic quantification of crack patterns by image processing. Comput. Geosci. 57, 77–80. https://doi.org/10.1016/j.cageo.2013.04.008.

Liu, J., Tang, C., Shi, B., 2022. Rock and soil mechanics evolution of desiccation cracking behavior of clays under drying- wetting cycles. Rock Soil Mech. 42, 2763–2772.
Liu, Y., Zhao, Y., Vanapalli, S.K., Mehmood, M., 2024. Soil-water characteristic curve of expansive soils considering cumulative damage effects of wetting and drying cycles. Eng. Geol. 339, 107642. https://doi.org/10.1016/j.enggeo.2024.107642.
Louati, F., Trabelsi, H., Jamei, M., Taibi, S., 2021. Impact of wetting-drying cycles and cracks on the permeability of compacted clayey soil. Eur. J. Environ. Civ. Eng. 25, 696–721. https://doi.org/10.1080/19648189.2018.1541144.
Mahannopkul, K., Jotisankasa, A., 2019. ScienceDirect Influences of root concentration and suction on *Chrysopogon zizanioides* reinforcement of soil. Soils Found. 59, 500–516. https://doi.org/10.1016/j.sandf.2018.12.014.
Meteorological Department of Thailand, 2020. Weather Forecast [WWW Document]. URL. https://www.tmd.go.th/en/.
Ng, C.W.W., Kamchoom, V., Leung, A.K., 2016. Centrifuge modelling of the effects of root geometry on transpiration-induced suction and stability of vegetated slopes. Landslides 13, 925–938. https://doi.org/10.1007/s10346-015-0645-7.
Nguyen, B.T., Ishikawa, T., Murakami, T., 2020. Effects evaluation of grass age on hydraulic properties of coarse-grained soil. Transp. Geotech. 25, 100401. https://doi.org/10.1016/j.trgeo.2020.100401.
Phan, T.N., Likitlersuang, S., Kamchoom, V., Leung, A.K., 2021. Root biomechanical properties of *Chrysopogon zizanioides* and Chrysopogon nemoralis for soil reinforcement and slope stabilisation. L. Degrad. Dev. 32, 4624–4636. https://doi.org/10.1002/ldr.4063.
Phan, T.N., Leung, A.K., Kamchoom, V., Likitlersuang, S., 2022. Reinforcement losses in soil stabilisation due to decomposing roots of *Chrysopogon zizanioides* and Chrysopogon nemoralis. L. Degrad. Dev. 34, 1080–1096.
Piroird, K., Lazarus, V., Gauthier, G., Lesaine, A., Bonamy, D., Rountree, C.L., 2016. Role of evaporation rate on the particle organization and crack patterns obtained by drying a colloidal layer. Europhys. Lett. 113, 38002. https://doi.org/10.1209/0295-5075/113/38002.
Rahardjo, H., Satyanaga, A., Leong, E.C., Santoso, V.A., Ng, Y.S., 2014. Performance of an instrumented slope covered with shrubs and deep-rooted grass. Soils Found. 54, 417–425. https://doi.org/10.1016/j.sandf.2014.04.010.
Rajamanthri, K., Jotisankasa, A., Aramrak, S., 2021. Effects of *Chrysopogon zizanioides* root biomass and plant age on hydro-mechanical behavior of root-permeated soils. Int. J. Geosynth. Gr. Eng. 7, 1–13. https://doi.org/10.1007/s40891-021-00271-0.
Richardson, A.E., Barea, J.M., McNeill, A.M., Prigent-Combaret, C., 2009. Acquisition of phosphorus and nitrogen in the rhizosphere and plant growth promotion by microorganisms. Plant Soil 321, 305–339. https://doi.org/10.1007/s11104-009-9895-2.
Sadeghi, H., Jabbarzadeh, M., Tourchi, S., 2024. Thermo-hydro-mechanical modelling of the heterogeneous subsidence and swelling in the desiccation cracked clayey strata. Eng. Geol. 343. https://doi.org/10.1016/j.enggeo.2024.107798.
Sorensen, J.N., 2000. Ontogenetic changes in macro nutrient composition of leaf-vegetable crops in relation to plant nitrogen status: a review. J. Veg. Crop. Prod. 6, 75–96. https://doi.org/10.1300/J068v06n01_08.
Stokes, A., Norris, J.E., Van Beek, L.P.H., Bogaard, T., Cammeraat, E., Mickovski, S.B., Jenner, A., Di Iorio, A., Fourcaud, T., 2008. How vegetation reinforces soil on slopes. Slope Stab. Eros. Control Ecotechnol. Solut. 65–118.
Stokes, A., Norris, J., va Beek, L.P.H., Boograd, T., Cammeraat, E., Mickovski, S., Jenner, A., Iorio, A., Fourcaud, T., 2008a. Slope stability and erosion control: ecotechnological solutions. In: Slope Stability and Erosion Control: Ecotechnological Solutions. Springer, pp. 65–118.
Tang, C., Shi, B., Liu, C., Gao, L., Inyang, H.I., Asce, M., 2011. Experimental Investigation of the Desiccation Cracking Behavior of Soil Layers during Drying, pp. 873–878. https://doi.org/10.1061/(ASCE)MT.1943-5533.0000242.
Tang, C.S., Cheng, Q., Leng, T., Shi, B., Zeng, H., Inyang, H.I., 2020. Effects of wetting-drying cycles and desiccation cracks on mechanical behavior of an unsaturated soil. Catena 194, 104721. https://doi.org/10.1016/j.catena.2020.104721.
Tian, B.G., Cheng, Q., Tang, C.S., Shi, B., 2023. Healing behaviour of desiccation cracks in a clayey soil subjected to different wetting rates. Eng. Geol. 313, 106973. https://doi.org/10.1016/j.enggeo.2022.106973.
Troung, P., Van, T.T., Pinners, E., 2008. The Vetiver Sistem for Slope Stabbilization, 1st ed. The Vetiver Network International.
Tsiampousi, A., Zdravkovic, L., Potts, D.M., 2017. Numerical study of the effect of soil–atmosphere interaction on the stability and serviceability of cut slopes in London clay. Can. Geotech. J. 54, 405–418. https://doi.org/10.1139/cgj-2016-0319.
Verbist, K., Torfs, S., Cornelis, W.M., Oyarzún, R., Soto, G., Gabriels, D., 2010. Comparison of single- and double-ring infiltrometer methods on stony soils. Vadose Zo. J. 9, 462–475. https://doi.org/10.2136/vzj2009.0058.
Wang, G., Huang, Y., Li, R., Chang, J., Fu, J., 2020. Influence of Vetiver Root System on Mechanical Performance of Expansive Soil: Experimental Studies 2020.
Wasino, R., Likitlersuang, S., Janjaroen, D., 2019. The performance of vetivers (Chrysopogon zizaniodes and Chrysopogon nemoralis) on heavy metals phytoremediation: laboratory investigation. Int. J. Phytoremed. 21, 624–633. https://doi.org/10.1080/15226514.2018.1546275.
Wei, L., Yao, P., Jing, G., Ye, X., Cheng, J., 2019. Root production, mortality and turnover in soil profiles as affected by clipping in a temperate grassland on the Loess Plateau. J. Plant Ecol. 12, 1059–1072. https://doi.org/10.1093/jpe/rtz039.
White, I., Sully, M.J., 1987. Macroscopic and microscopic capillary length and time scales from field infiltration. Water Resour. Res. 23, 1514–1522. https://doi.org/10.1029/WR023i008p01514.
Wu, Z., Leung, A.K., Boldrin, D., Ganesan, S.P., 2021. Variability in root biomechanics of *Chrysopogon zizanioides* for soil eco-engineering solutions. Sci. Total Environ. 776, 145943. https://doi.org/10.1016/j.scitotenv.2021.145943.
Xu, Y., Guo, Y., Huang, Z., Liu, D., Huang, Q., Tang, H., 2023. Study on *Cynodon dactylon* root system affecting dry–wet cracking behavior and shear strength characteristics of expansive soil. Sci. Rep. 13, 1–17. https://doi.org/10.1038/s41598-023-39770-7.
Yang, Z.-M., Cheng, Q., Tang, C.-S., Tian, B.-G., Zhang, X.-Y., 2023. Coupled effects of temperature and relative humidity on desiccation curling of a clayey soil. Eng. Geol. 327, 107370. https://doi.org/10.1016/j.enggeo.2023.107370.
Yong, R.N., Warkentin, B.P., 1975. Soil Properties and Behavior. Elsevier, Amsterdam.
Yu, Z., Eminue, O.O., Stirling, R., Davie, C., Glendinning, S., 2021. Desiccation cracking at field scale on a vegetated infrastructure embankment. Geotech. Lett. 11, 88–95. https://doi.org/10.1680/jgele.20.00108.
Yuliana, Y., Apriyono, A., Leung, A.K., Keawsawasvong, S., Kamchoom, V., 2023. The impact of tree transpiration on the safety and serviceability of pavement under seasonal variations. Int. J. Pavement Eng. https://doi.org/10.1080/10298436.2023.2283112.
Zhang, Y., Niu, J., Zhang, M., Xiao, Z., Zhu, W., 2017. Interaction between plant roots and soil water flow in response to preferential flow paths in Northern China. L. Degrad. Dev. 28, 648–663. https://doi.org/10.1002/ldr.2592.
Zhu, P., Zhang, G., Wang, H., Xing, S., 2020. Soil infiltration properties affected by typical plant communities on steep gully slopes on the Loess Plateau of China. J. Hydrol. 590, 125535. https://doi.org/10.1016/j.jhydrol.2020.125535.